

RH124 – RedHat Administrator Level 1

Chapter 1 - Getting Started with Red Hat Enterprise Linux

What is Linux ?

- Linux is an **operating system kernel** that, combined with GNU and other components, forms complete operating systems used on servers, desktops, phones, and embedded devices worldwide.
- It is developed collaboratively as **open source**, allowing anyone to **view, modify, and distribute** the code under the GPLv2 license, which enabled rapid growth and adoption.
- Linux is a free, Unix-like operating system kernel created by Linus Torvalds in 1991 to build a personal, affordable Unix for his 386 PC and to learn operating systems by doing, which later grew into a global open-source project under the GNU GPL license.
- **Linus Torvalds**, then a computer science student at the University of Helsinki, started the Linux kernel and continues to oversee its development as the final authority on mainline kernel changes.
- He later worked at Transmeta and then under the Open Source Development Labs, which merged into today's Linux Foundation, while remaining the project's central maintainer.

What is RedHat Linux ?

- RedHat is an **open-source** software company best known for **Red Hat Enterprise Linux (RHEL)**, a commercially supported Linux platform used widely in **enterprises, clouds, and datacenters**. It began in the mid-1990s around a polished Linux distribution and a business model of subscriptions for **support, security, and lifecycle management**.
- The project evolved from boxed "**Red Hat Linux**" into two streams in 2003: community-driven **Fedora** and **enterprise-grade RHEL**, aligning innovation upstream with long-term support downstream.

What Red Hat provides ?

- The flagship is RHEL, **offering long-term support, certified hardware/software ecosystems, SELinux-based security**, and predictable lifecycle for enterprises.
- Beyond the OS, the portfolio spans **automation (Ansible), containers and Kubernetes platforms (OpenShift), middleware, storage, virtualization, and edge solutions** to run workloads consistently across on-prem and cloud.

Where it's used today ?

- Enterprise servers and databases in **corporate and government datacenters** for **reliability, security updates**, and vendor certifications.
- **Public cloud and containers** as the base OS for cloud instances and container hosts, often paired with **OpenShift for Kubernetes operations**
- High-performance computing, telecom, and financial services where **stability, performance, and compliance are critical**.

Linux Flavors

- Ubuntu
- Debian
- Fedora
- Red Hat Enterprise Linux (RHEL)
- CentOS Stream
- Arch Linux
- Linux Mint
- Kali Linux
- Parrot OS
- Raspberry Pi OS

Windows Vs linux		
Satyender Sharma Tech Coach		
Feature	Windows	Linux
User Interface	Focused on GUI, uniform appearance across versions.	Fully customizable to user preferences.
Cost	Licensed OS with multiple versions (Home, Pro).	free and paid, open-source, various distributions (Debian, CentOS)
Software	Extensive range of paid and free apps.	Open-source programs readily accessible.
Compatibility	Broad hardware and software support.	Varies by distribution and hardware.
Security	Prone to malware and viruses.	Lower risk due to limited user base.
Customization	Fewer options compared to Linux.	Maximum customization available.
Updates	Automatic updates with frequent patches.	Managed at the distribution level.
Performance	Generally good performance.	Lightweight on resources, and is best for older hardware.
Support	Comprehensive support from Microsoft and third parties.	Active community, forums, and guides.

